Rady | School of Management
UNIVERSITY OF CALIFORNIA SAN DIEGO

MGTA 453: Business Analytics

STUDENT CLASS (2027)
TERM (FA26)

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DESCRIPTION

In today's increasingly global and competitive economy, businesses are striving to use analytics (i.e., the use of data, together with statistical and quantitative models, to make better, data-driven business decisions) to gain competitive advantage. Yet, there is still a significant shortage of business analysts and project managers who are adept at effectively utilizing analytics to create substantial value to an organization by solving impactful business problems.

The use of business analytics has exploded and become pervasive in small and large businesses. Business analytics now drives ticket pricing, retail assortments, movie recommendations, digital marketing campaigns, healthcare scheduling, crime fighting, social media investments, capacity decisions and more. Today, companies generate an immense volume of transactional data, capturing trillions of bytes of information about their customers, suppliers, and operations. Millions of networked sensors are embedded in the physical world in devices such as computers, mobile phones, smart energy meters, automobiles and industrial machines that sense, create, and communicate data. In a digitized world, consumers communicate, browse, buy, share, and search, creating extensive trails of valuable data.

In this course, students are introduced to the foundations of business analytics. The course is designed to teach business managers how to use *data* to make good *decisions* in complex decision-making situations. In fact, a good *decision* is not the same as a good *outcome*. A good outcome can sometimes result as a matter of luck, and, conversely, a bad outcome does not necessarily suggest that a manager has made a bad decision. Students will develop sound reasoning skills and learn how to utilize information to arrive at good decisions.

OBJECTIVES

To be an intelligent user of analytics, a business analytics professional must be able to:

- Identify an underlying analytical structure in a seemingly complex and amorphous decision problem
- Understand the role of uncertainty and risk in the decision-making process
- Analyze available data to understand relationships among variables and to create predictions
- Understand the trade-offs involved in the decision
- Use available computing technology (e.g., R, Radiant, Python, and so on) to arrive at optimal solutions; students will gain experience programming in R and Python and using relevant libraries

To equip you with these skills, several analytical methods will be covered including:

- Decision analysis
- Statistical inference
- Simulation modeling
- Regression analysis
- Optimization

MATERIALS

Required

- *Data, Models, and Decisions: The Fundamentals of Management Science*, by Dimitris Bertsimas and Robert M. Freund, 2004, Dynamic Ideas, Belmont, MA, ISBN 0-9759146-0-X.

COURSE SCHEDULE

Week	Topic(s)	Due
1	Decision analysis	
2	Statistical inference	Assignment #1 [C], Case: Investor's Dilemma [G]
3	Statistical inference Simulation	Assignment #2 [C]
4	Simulation	Assignment #3 [C], Case: Shared Social Responsibility [G]
5	Advanced Simulation Optimization	Case: Ontario Gateway [G]
6	Regression analysis	Assignment #4 [C]
7	**Midterm Exam**	Midterm Exam [I]
8	Regression analysis	Assignment #5 [C], Case: Hot Hand [G]
9	Optimization	Assignment #6 [C], Case: Croq' Pain [G]
10	Optimization Regression analysis	Assignment #7 [C], Case: Filatoi Riuniti [G]
11	**Final Exam**	Final Exam [I]

ASSIGNMENTS & DELIVERABLES

All assignments will be clearly marked in Canvas with both a due date and a specification on the extent of collaboration that is permitted:

Key

- I – Independent, individual work only. No collaboration or consultation allowed.
- G – Students may work together only within their own study groups and turn in one deliverable for the entire group
- C – Collaboration with classmates is allowed. However, each student must submit a separately written up or produced deliverable (not a copy)

Students are expected to adhere to these guidelines. The referencing of assignment solutions for this class from prior years and/or those available on the internet is strictly prohibited.

Case Studies

There will be six case write-ups. One report per team should be submitted electronically by the **beginning of class** on the day we cover each case. It is recommended that you keep a copy for your own reference during class. The goal of each case study is for you to discuss the case in detail with your team and then together agree upon recommendations. These recommendations should be backed up by solid arguments, evidence, and mathematical or data analysis as needed.

A case write-up should consist of (i) an executive memo that is no more than two single-sided pages of text, and (ii) up to six pages of supporting documents (charts, figures, calculations, etc.). The memo should not have any calculations and be written in a managerial style that clearly articulates your recommendations.

Exams

There will be two exams in the class. The midterm exam will occur during the 7th week of the quarter, and the final exam will occur during the week following the final class session (i.e., the 11th week). For both exams, you will have 3 hours to complete the exam. The format of the exam will be communicated prior to the exam taking place. You may not communicate with anyone or seek help from any third party during exams. You may not access your phone or use AI tools during exams.

Class Participation

Your class participation grade is based upon your contribution to class discussion, case presentations, group work, familiarity with required readings, and the insights reflected in your responses to classroom questions. To prepare for class, you should fully complete the required assignments prior to each session. You should strive to be an active participant during class and contribute to the quality of the discussion. Please note that the frequency of your interventions in class is not a key criterion for effective class contribution. On the other hand, attendance is a factor in your ability to contribute to the learning environment. Peer evaluations from study group members is also a factor. Based on these criteria, your class participation grade will be assigned subjectively between 0 and 15 points.

GRADING

Exams and Deliverables	Percentage
Final Exam	35%
Midterm Exam	25%

Case Write-Ups	15%
Homework Assignments	10%
Class Participation	15%
Total	100%

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2

ACADEMIC INTEGRITY VIOLATIONS

If you are found responsible for an academic integrity violation after going through the Academic Integrity Office's (AIO) process, then in addition to the administrative sanctions you receive from the university (https://academicintegrity.ucsd.edu/process/consequences/sanctioning-guidelines.html), you will receive academic consequences as well in the course. I strictly follow the faculty guidance on assessing academic consequences which can be found here: https://academicintegrity.ucsd.edu/_files/process-files/Faculty-to-Faculty%20Advice%20on%20Academic%20Sanctions_Handout_PDF.pdf. Be sure to note that "Exam Cheating, Falsification/Fabrication/Fraud or Contract Cheating" will result in your receiving an "F" in the course, and thus you will be required to retake the course next year. The academic consequences are non-negotiable and do not depend on your particular circumstances (e.g., concerns about your fellowship, a company sponsoring you in the program, graduating on time, etc.). While I appreciate these are very real concerns, it is also easy to avoid by maintaining your academic integrity. It is much better to receive an average or poor grade, then to receive an "F", delay your graduation, and even possibly being removed from the program. I strongly encourage you to make smart academic integrity decisions in this course as well as your entire time here at UC San Diego.

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or **osd@ucsd.edu.**

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CLASS POLICY ON THE USE OF ARTIFICIAL INTELLIGENCE TOOLS

The use of artificial intelligence (AI) tools such as ChatGPT, Bing Chat, Bard, Claude, etc. is encouraged in this class. These tools have considerable potential to streamline and augment our work. You should use them with the following caveats and conditions:

- AI-tools should enhance your work, not replace it. For instance, large language models (LLMs) can improve the quality of your writing, identify new sources, provide high-level summaries of complex ideas or terms, generate code, and more. These tools can help you come up with new ideas, overcome writer's block, and refine your original ideas. You should not use Generative AI tools to generate content that you simply put your name on. This constitutes an academic integrity violation, much like taking another individual's work and misrepresenting it as your own.

- You may utilize Generative AI and LLMs for class preparation and for help with assignments. However, we require that you fully understand and can explain, in-person or on video, all details of every line of code and mathematical calculation that you submit as your own work.

- If you use AI for an assignment, be transparent about how it was used in a footnote or citation (e.g., a link to the ChatGPT chat that you created).

- Be aware that LLMs have documented issues with hallucination and inaccuracy (e.g., generating fake python packages, failing logical arguments, and misrepresenting sources). You are fully responsible for all content you submit in your assignments and will be graded accordingly.